

A learners' microcomputer Part 2

How to get into microcomputing without boiling your brain cells or breaking the bank

This month we go into the 'architecture' of the micro-processor, an RCA CDP1802, detailing the functions of the various portions necessary for an understanding of how our computer works.

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Development: **Graeme Teesdale**

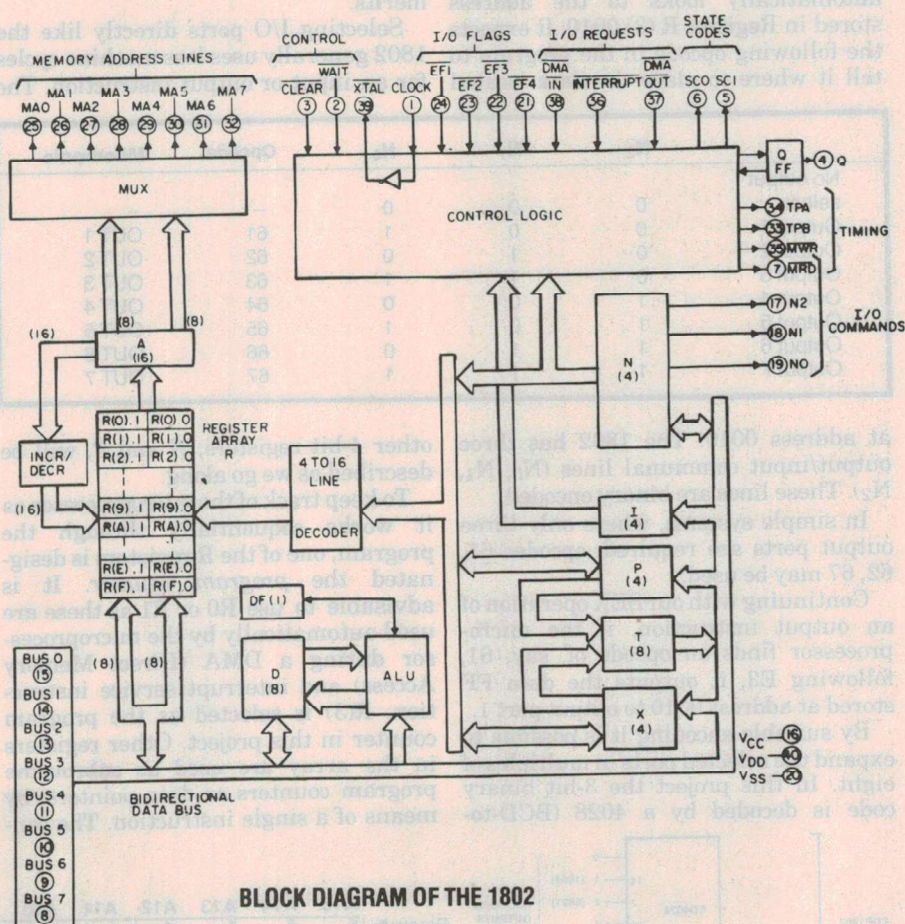
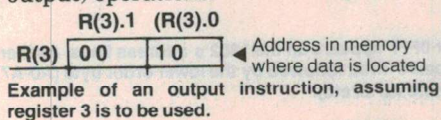
THE OBJECTIVE of this project is to teach the beginner something about microprocessors while attempting the practical aspect of it. The following text in general will relate to devices used in this project; however, some comparisons with other device groups will be done.

Architecture

A block diagram of the 1802 is illustrated here.

The principal feature of this microprocessor is the sixteen 16-bit register arrays, labelled (R). The 16-bit register gives an addressing capability of 2^{16} , i.e: 65 536 bytes, normally known as a memory system of 64K bytes. Because the 1802 is only an 8-bit microprocessor it is necessary to handle the 'R' register in two 8-bit bytes, designated as R(x).1 for high order bytes and R(x).0 for low order bytes, where R(x).1 indicates the selected 'R' register D to F, i.e: 0-15, and registers over decimal 9 are indicated in a hexadecimal code using letters A to F. The array can also be used as a 32 by 8-bit array if necessary.

Individual registers in the array (R) are selected by one of three 4-bit binary code registers: N, P, X — these in turn are converted to a 16-bit address to the R registers by the 4-to-16 line decoder. Each of the three registers has a particular use. For example: the X designator selects one of the 16 registers R(x) to 'point' to the memory for an operand (or data) in certain ALU (Arithmetic Logic Unit) or I/O (input-output) operations.



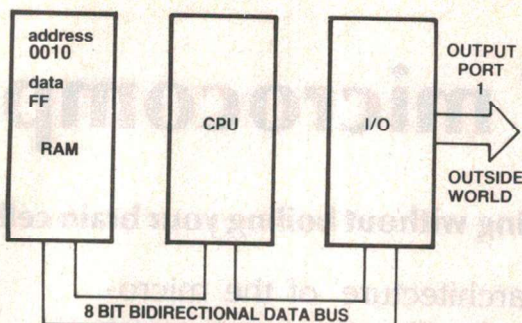
The memory system can be considered as a number of pigeon holes (up to 64K) starting at address 0000 and going to FFFF, each having eight individual storage compartments. I.e: an 8-bit data word (or byte) can be stored in two hexadecimal numbers, and hence

the data at memory location 0010 could be, for example, FF.

An output instruction will take the data FF stored in location 0010 and place it at an output port connected to the outside world.

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How the microprocessor (CPU), random access memory (RAM) and input/output (I/O) sections of the computer are interconnected.



In this case the data is stored in RAM, memory which can be written to or read from.

The control instruction for the 'x' register is SET X (mnemonic SEX!) and opcode of EN where N is the selected 'R' register. Getting confused yet?! As we have selected Register 3, opcode is E3. When the microprocessor reads the opcode E3 in the program it automatically looks to the address stored in Register R (3) 0010. It expects the following opcode in the program to tell it where to place the data located

decimal decoder) to enable selection of RAM, a colour encoder IC and I/O IC plus additional expansion ports.

Not all microprocessors use this system; some consider designated memory locations as an I/O port. An example is the Motorola 6800 microprocessor. The popular Motorola 6800 D2 kit considers I/O ports at addresses 8004, 8006. Both methods have their merits.

Selecting I/O ports directly like the 1802 generally uses less machine cycles for an input or output instruction. The

	N ₂	N ₁	N ₀	Opcode	Mnemonic
No output select	0	0	0	—	—
Output 1	0	0	1	61	OUT 1
Output 2	0	1	0	62	OUT 2
Output 3	0	1	1	63	OUT 3
Output 4	1	0	0	64	OUT 4
Output 5	1	0	1	65	OUT 5
Output 6	1	1	0	66	OUT 6
Output 7	1	1	1	67	OUT 7

at address 0010. The 1802 has three output/input communal lines (N₀, N₁, N₂). These lines are binary encoded.

In simple systems, where only three output ports are required, opcodes 61, 62, 67 may be used.

Continuing with our SEX operation of an output instruction, if the microprocessor finds an opcode of, say, 61, following E3, it outputs the data FF stored at address 0010 to output port 1.

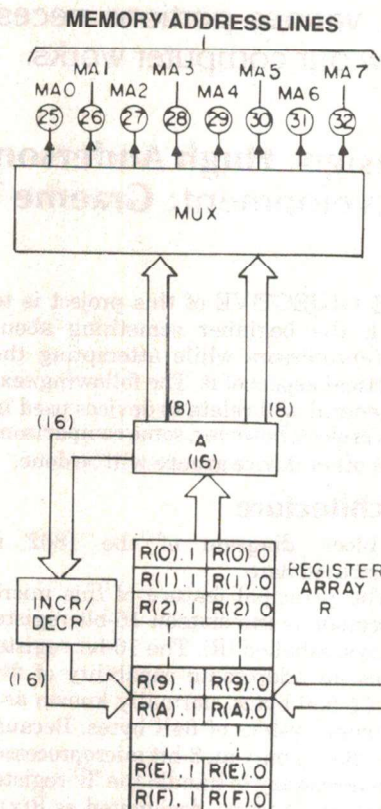
By suitable encoding it is possible to expand the selected ports in multiples of eight. In this project the 3-bit binary code is decoded by a 4028 (BCD-to-

other 4-bit registers, N and P, will be described as we go along.

To keep track of the microprocessor as it works sequentially through the program, one of the R registers is designated the *program counter*. It is advisable to use R0 or R1 as these are used automatically by the microprocessor during a DMA (Direct Memory Access) and interrupt service instruction. R(3) is selected as the program counter in this project. Other registers in the array are used as subroutine program counters or data pointers, by means of a single instruction. The con-

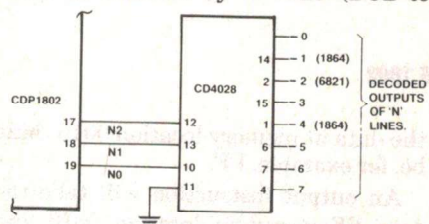
tents of the P register can be used to call any R registers. The control instruction is SET P (mnemonic SEP), opcode DN, where once again N is the designated 'R' register.

To enable the microprocessor to select any address location from 0000 to FFFF a 16-bit address buss must be used. To reduce the pinout count on the 1802 microprocessor chip, RCA have a bit-multiplexed address line labelled MA0 to MA7.

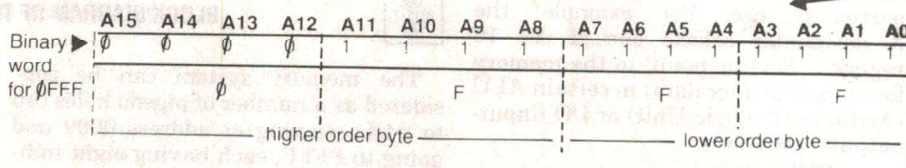


In each cycle, the high order byte, A8 — A15, appears on the memory address lines MA0-7 first. Those bits required by the memory address are latched by a timing pulse, TPA. The low-order byte of the 16-bit address appears on the address lines after the termination of TPA. Latching of all eight higher-order address bits would permit a memory system of 64K bytes.

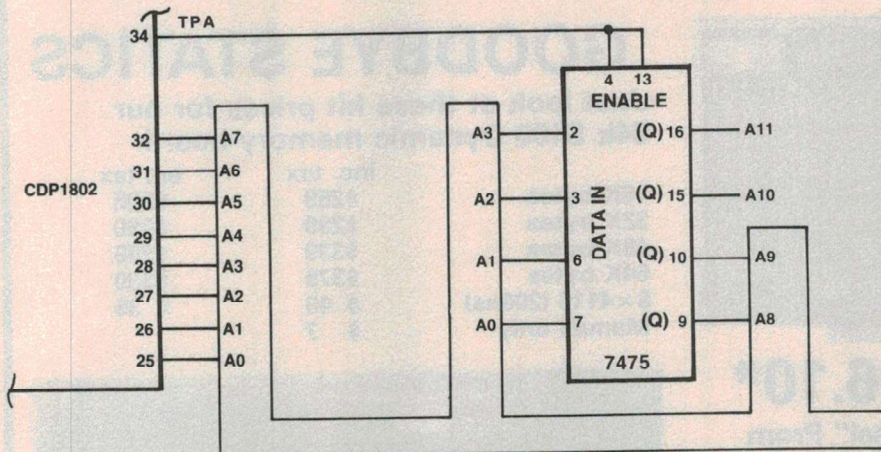
Taking address 0FFF as an example:



The 4028 BCD-to-decimal decoder decodes the CPU's 3-bit I/O commands and selects a colour encoder IC (1864), an I/O IC (6821) or RAM.



How the binary word represented by the hex number 0FFF appears on the 1802's address lines. As there are only eight lines, the higher order byte (A8-A15) appears first, followed by the lower order byte (A0-A7). In the 1802, the order of the lines is inverted (A8 is where A0 is, etc).



Part of the latch circuit used to temporarily store the lower four bits of the higher order address byte. When the TPA pulse from the 1802 goes low, the 7475 puts the data that was present on the data inputs, at the time TPA was present, onto address lines A8-11 while the 1802 switches the lower order address byte onto the data buss lines A0 to A7.

It can be observed from the table at lower left that it is necessary to decode only the lower four bits of the higher order byte plus all of the lower order bytes for the microprocessor to address memory location 0FFF. It is important to remember that with the 1802 multi-

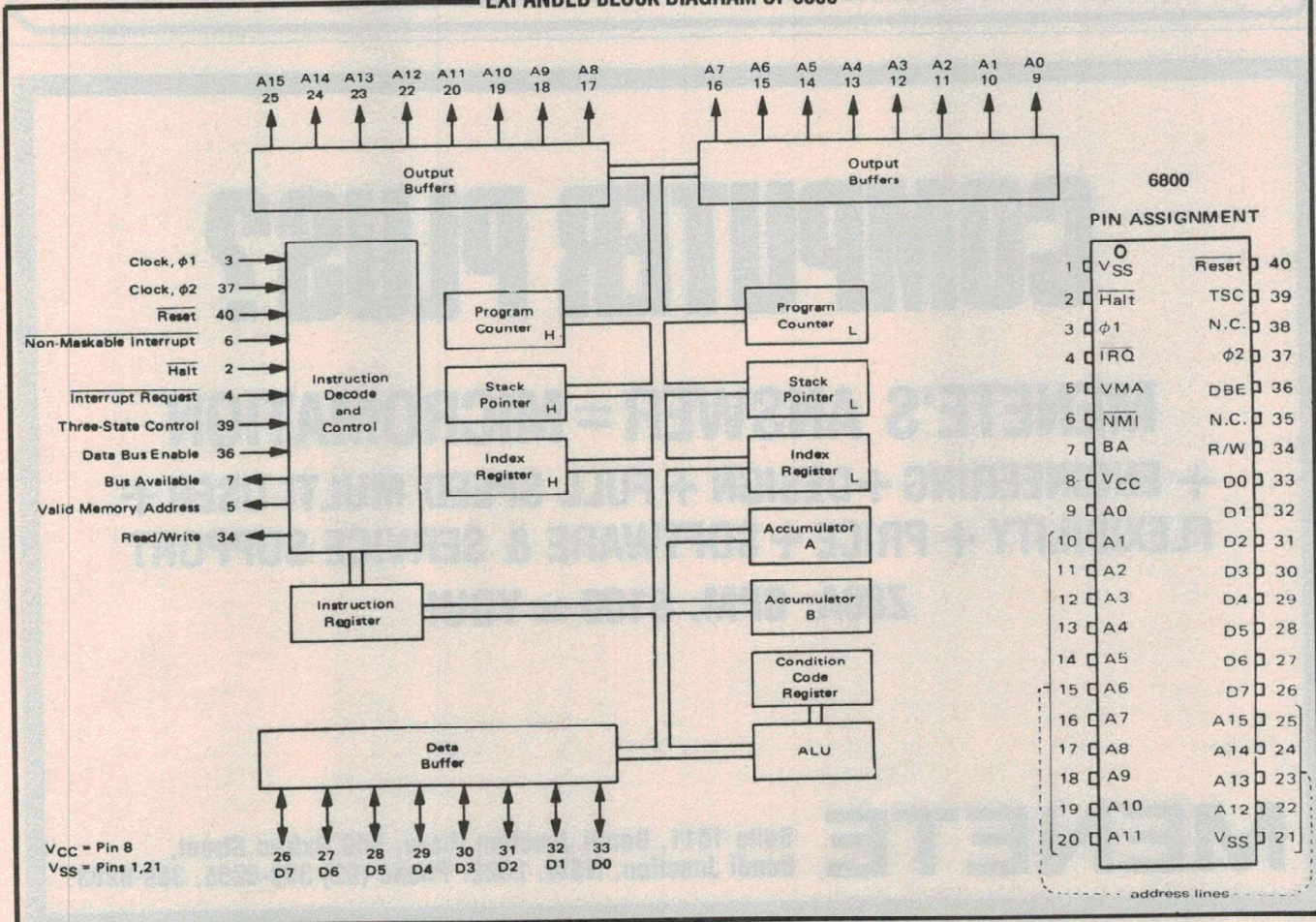
plexed address lines A8 becomes A0 and so on, up to A15 (i.e: they're inverted).

The 7475 is a 4-bit bistable latch and is used as temporary storage for the lower 4-bits of the high byte. When the ENABLE pins (4 and 13) go low, the information that was present at the data

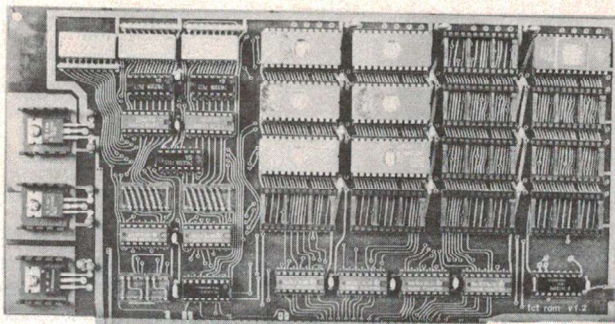
input at time of transition is returned at the Q outputs. The latching pulse (TPA) is obtained from the 1802 at pin 34. Just using this one 4-bit latch it is possible to decode the necessary address lines.

Some microprocessors have all sixteen address lines available: ►

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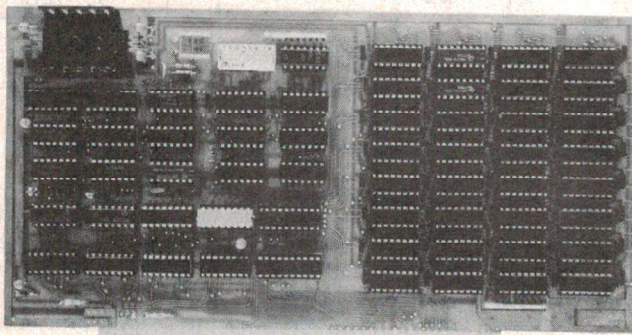
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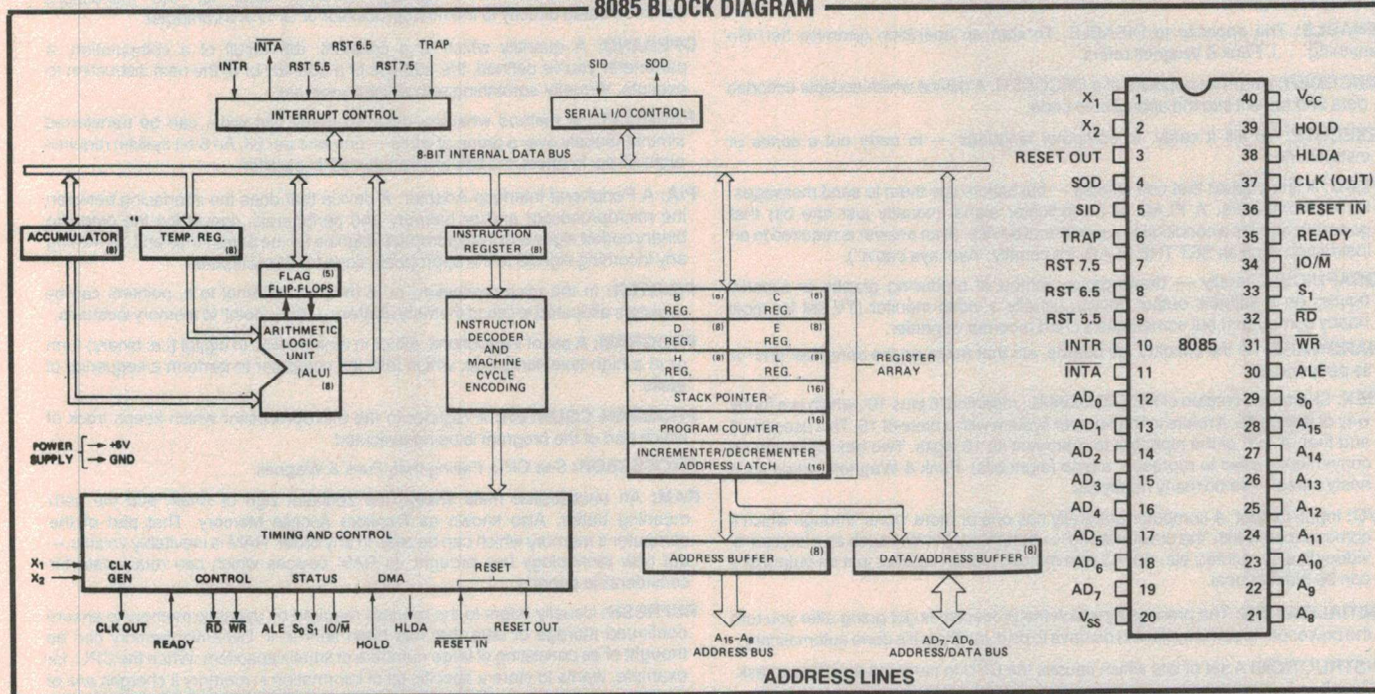
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The 8085 microprocessor has the lower order address bytes multiplexed on the data buss lines.

NOTE: Owing to a last-minute problem with supply of some critical components, we have had to hold over the description of the construction of the ETI-660 computer,

which should now commence in the July issue, assuming no further intervention of problems beyond our control, fairies at the bottom of the darkroom, etc.

8085 BLOCK DIAGRAM



☆☆☆ GLOSSARY OF TERMS ☆☆☆

ALU: Arithmetic Logic Unit. That portion of the microprocessor which performs calculations according to a set of instructions.

ADDRESS: A label or name (usually a binary or hexadecimal number) specifying a particular memory location.

ALPHANUMERIC: Letters and numbers.

ARRAY: A group of related variables or constants — generally given a name or a label. Arrays are often located in consecutive memory locations.

ASSEMBLER: A program which takes a series of mnemonics and puts out a corresponding binary code.

ASYNCHRONOUS: Refers to a circuit in which the various elements are not arranged to change state in synchronism, or in synchronism with another signal.

BCD: Binary-coded decimal. A binary code to represent decimal numbers; for example, a 4-bit binary code can represent the numbers 0 to 15.

BASIC: The most widely used high-level computing language in the hobbyist field. Allows you to communicate with a computer in a fair approximation of pidgin English.

BAUD: A measure of the speed or rate of data transmission. A baud is the transmission of one bit per second.

BINARY: The 'base two' number system in which everything is represented in 'ones' and 'zeros'.

BIT: A single binary digit, representing either a 'one' or a 'zero'.

BOOTSTRAP: This refers to the process of inserting instructions or entire routines directly onto the data and address busses of a computer.

BRANCH: A certain instruction included in a program which makes the processor perform a step out of the usual sequence, usually if a certain specified condition is satisfied. A branch instruction will skip or jump following instructions.

BUSS: Generally, a system of interwiring in which each line has a designated task or carries only specific signals. A buss permits communication between devices making up a computer and can be one-way (unidirectional) or two-way (bidirectional).

BYTE: A binary number, usually of eight bits. It can represent a number from 0 to 255 (8-bit byte) as there are 256 possible combinations of ones and zeros eight bits long.

CPU: Literally, 'Central Processing Unit'. Generally a shorthand term referring to the microprocessor in a microcomputer. One of those terms used more out of tradition than for the sake of clear writing.

CLOCK: An oscillator that provides timing signals which synchronise the operations carried out by the microcomputer.

DMA: Direct Memory Access (or Direct Memory Address) — means what it says. The term refers to the practice of fetching data directly from memory by an external (or peripheral) device without the need for intervention by the microprocessor. A good technique for speeding data transfer.

DEBUG: A BUG is a mistake or similar problem that has crept into a program. The act of debugging removes it. The terms were said to be introduced when a small insect crawled into an early computer and stuffed up its operation. The 'bug' was ceremoniously taped to a page of the report which explained the fault. No, we don't *quite* believe it either, but it's a good story anyway and its inclusion is a long-standing tradition.

DECODER: A device which changes one code to another. For example, a 4-bit binary code may be changed to a 1-of-16 code. Certain ICs are made to perform such a function, generally called 4-to-16 line decoders. A 4-bit to 1-of-10 decoder is called a BCD to decimal decoder — for each 4-bit code in, the decoder will activate the appropriate one of its ten output lines.

DISABLE: The opposite to ENABLE. To halt an operation — to turn something off (if only temporarily). Also, look it up in your Funk & Wagnell.

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DUMP: If you have a wheelbarrow full of dirt and tip it out, that's dumping the dirt. If you have your computer memory full of data and you transfer it to tape, that is called dumping. Note you can dump from memory to tape and from tape to memory. Pity you can't do it with wheelbarrows sometimes.

EPROM: Erasable Programmable Read Only Memory. First, look up ROM. It says, basically, that ROM is a memory you can't muck around with. An EPROM is thus a ROM you can rewrite — Erasable (the ability to rub out) and Programmable (the ability to program).

ENABLE: The opposite to DISABLE. To start an operation (promise her/him anything . . .). Funk & Wagnell refers.

ENCODER: Aha! The opposite of a DECODER. A device which accepts uncoded data and turns it into the appropriate code.

EXECUTE: To kill a cutey. In computer language — to carry out a series of instructions.

FLAG: A linen object that one salutes — but sailors use them to send messages. So do computers. A FLAG is an indicator signal (usually just one bit) that generally signals a condition to a peripheral device. (If an answer is required to an instruction such as SET THE FLAG, it's usually "Aye, aye cap'n").

GRAPHICS: Literally — drawings; a method of producing graphs or pictorial figures on a suitable output device, usually a video monitor (TV set for most hobby computers) but sometimes a chart recorder or printer.

HARDWARE: All the circuitry, pc boards, etc that make up the computer and/or its peripherals.

HEX: Shortened version of HEXADECIMAL, meaning '6 plus 10', which is a funny way of saying 16. It refers to the number system with a base of 16. This uses 0 to 9 and then A to F of the alphabet to represent its 16 digits. Two hex digits can be conveniently used to represent a byte (eight bits). Funk & Wagnell reckon it's a nasty curse — so do many hobbyists.

I/O: Input/Output. A computer generally has one or more 'ports' through which it communicates with 'the outside world' — peripheral devices such as a keyboard, video display, printer, etc. An I/O port may be just an input or just an output or it can be bidirectional.

INITIALISATION: The process by which the processor is got going after you turn the power on. In some systems you have to do it, in others it's done automatically.

INSTRUCTION: A set of bits which causes the CPU to carry out a particular task. Usually a basic or fundamental command understood by the microprocessor.

INSTRUCTION SET: That set of fundamental instructions which controls a microprocessor's or computer's basic set of possible operations. In general, the larger the instruction set the more powerful the microprocessor.

INTERFACE: A device or circuit which effects the transfer of data (in the appropriate form) from one system to another or one part of a system to another part. In homo sapien terms, this is represented by the FIST — as in, "ya wanna punch in ter face?"

INTERPRETER: A program which permits the computer to operate with a high-level language. It literally interprets what you've written in the high-level language into a code (bloody great strings of 1s and 0s) which the microprocessor 'understands' (i.e. acts on).

INTERRUPT: What happened to Mt St. Helens. In a computer, an interrupt suspends (hence, interrupts) normal program execution while something else happens (usually urgent and involving a peripheral).

LANGUAGE: A repertory of instructions — symbols, expressions, etc, used to 'call up' the instructions or procedures a processor can execute. Higher-level languages are easily understood by mere humans and computers of all sorts designed to work with such languages (BASIC, for example) use an INTERPRETER to change this into the MACHINE CODE under which the processor operates.

LOAD: The opposite to DUMP. Generally it means 'to store' binary data, usually into some form of memory.

MACHINE CODE: The binary form of instructions which a processor actually 'understands'.

MEMORY: Sort of 'electronic pigeon holes' in which binary data is stored. There are many forms of memory, generally in two classes: volatile and non-volatile. Volatile memory 'loses' what's stored in it when you turn the power off. Memory ICs (RAM chips) are an example. Non-volatile memory permanently stores data. Cassette tape is an example.

MEMORY-MAPPED: An organisation of the hardware and software of a computer to enable data to be transferred directly from memory to an output port without involvement of the microprocessor.

MNEMONIC: Not related to manic, but you'll become that if you try and remember too many mnemonics. A mnemonic is a nickname for a particular instruction and is thus made easy for us humans to remember (we have quasi-volatile memories, generally affected by lack of sleep, drink and self-abuse). The 1802 microprocessor, for example, has one mnemonic called 'SEX' — see how easy mnemonics are to remember!

MODULATOR: A device, included in every good hobby computer, which takes the computer's output and converts it to an RF signal on a suitable TV channel, giving you a cheap, convenient visual display — even if you don't understand what's on the screen.

MULTIPLEX: The principle of arranging or communicating information from a number of sources by selecting each source sequentially.

OPCODE: Hexadecimal or binary numbers used for the instructions communicated directly to the microprocessor or CPU of a computer.

OPERAND: A quantity which is a constant, the result of a computation, a parameter you've defined, the address of a quantity or of the next instruction to execute. Virtually 'something with which to operate'.

PARALLEL: A method whereby data, so many bits wide, can be transferred simultaneously over a group of wires — one wire per bit. An 8-bit system requires eight wires. In effect, the bits are transferred 'in parallel'.

PIA: A Peripheral Interface Adaptor. A device that does the interfacing between the microprocessor and/or memory and peripherals, converting the outgoing binary coded signals to the appropriate signals for the peripherals and converting any incoming signals to the appropriate code for the computer.

POINTER: In the microprocessor, or in memory external to it, pointers can be registers allocated to listing memory address — they 'point' to memory locations.

PROGRAM: A set of instructions, either in mnemonics, in digital (i.e. binary) form or in a high-level language, which tells the computer to perform a sequence of tasks.

PROGRAM COUNTER: A register in the microprocessor which keeps track of which part of the program is being executed.

PROCESSOR: See CPU. Failing that, Funk & Wagnell.

RAM: An uncastrated male sheep; the zodiacal sign of Aries, and the verb meaning batter. Also known as Random Access Memory. That part of the computer's memory which can be read in any order. RAM is inevitably volatile — but new technology has brought us RAM devices which can retain data for considerable periods.

REFRESH: Usually refers to the process required by dynamic memory to ensure continued storage of data that has been sent to it. Dynamic memory can be thought of as consisting of large numbers of small capacitors. When the CPU, for example, wants to store a specific bit of information in memory it charges one of these small capacitors. The capacitor will slowly discharge itself, however, so additional hardware is used to look periodically at all of the capacitors and refresh the charge on each one.

RESET: Simply — go back to the start, do not pass GO, do not collect 200 bytes.

REGISTER: A general-purpose memory, or set of memory locations, built into the microprocessor itself. Sometimes, particular registers may be designated for a specific purpose.

ROM: Read-Only Memory — the opposite to Write-Only Memory (the latter being fairly rare and thus unconscionably expensive for the hobbyist, apart from being totally useless). Read-Only Memory is memory which cannot be altered or otherwise mucked about with. In humans it is that part of the brain which stores forever shameful, embarrassing or otherwise traumatic events.

ROUTINE: A whole program or part of a program designed to perform a single function or action.

SCRATCHPAD: A piece of wood on which cats are supposed to sharpen their claws in lieu of the furniture. In a microprocessor, a scratchpad is an area of memory used to store data temporarily or for making quick data transfers. It is probably the most frequently used area of memory.

SERIAL: The sequential transfer of data — one bit after another. Usually only one wire is needed for serial data transfer.

SOFTWARE: Programs, routines, instructions, mnemonics — all that stuff the electronics works on.

STACK: A colloquial expression for a vehicle accident. Here, we're talking about a sequence of registers or memory locations. Usually, the last bit of data put in is the first fetched out.

STACK POINTER: An address that specifies the location of the last entry in the STACK.

STROBE: To rapidly read registers or memory locations in sequence.

SUBROUTINE: A part of a program which performs a specific task and which is available for use elsewhere in a program or routine, as often as you like.

VDU: Visual Display Unit. Usually a TV set in hobby computers, but may be a video monitor which accepts the digital output direct from the computer.

VARIABLE: A specified location in memory (usually RAM) which is allocated a specific meaning in a program or routine. It may contain a fixed data value or changing data during program execution.

£@%*!: Term used by extremely frustrated or provoked computer hobbyist when his program refuses to be debugged or his computer won't boot up.